LET'S DISCUSS...

ROOT CAUSE ANALYSIS

It's one of the key techniques in systems analysis!

CONTEXT

Where is root cause analysis within the broader systems analysis field?

PROBLEM

We need an IT problem to analyze

How about infinityfree.com problems we have had!

THE SOLUTION

Examine information systems analysis techniques

Choose a suitable one

Learn and apply to the problem

ROOT CAUSE ANALYSIS: WHAT IS IT?

WHY?

...because just covering up the symptoms might not be the best way

IF SOMETHING HURTS...

take a painkiller pill?

That might solve the problem

It might *not* solve the problem

Examples?

PATCH THE PROBLEM?

Car wheel about to fall off?

Wrap lots of tape around it

No - find out why, and fix that

ROOT CAUSE ANALYSIS:

Find the root of the problem

That enables fixing the underlying cause

CONTEXT: THE SYSTEMS ANALYSIS FIELD

SYSTEMS ANALYSIS

Specifying and designing improved organization IT processes

Improvement techniques include

PROCESS MAPPING

visualizations of existing processes (to identify possible improvements)

ROOT CAUSE ANALYSIS

finding underlying causes of problems (to identify what to improve)

BENCHMARKING

comparing organizational processes to other organizations' processes (to identify areas to improve)

PROTOTYPING

building throwaway test solutions to decide what to build for real (to identify how to do improvements)

COST-BENEFIT ANALYSIS

comparing the benefits of a process change to its costs (to decide if improvement is worth doing)

AUTOMATION

using technology to boost labor productivity (to improve process quality and cost)

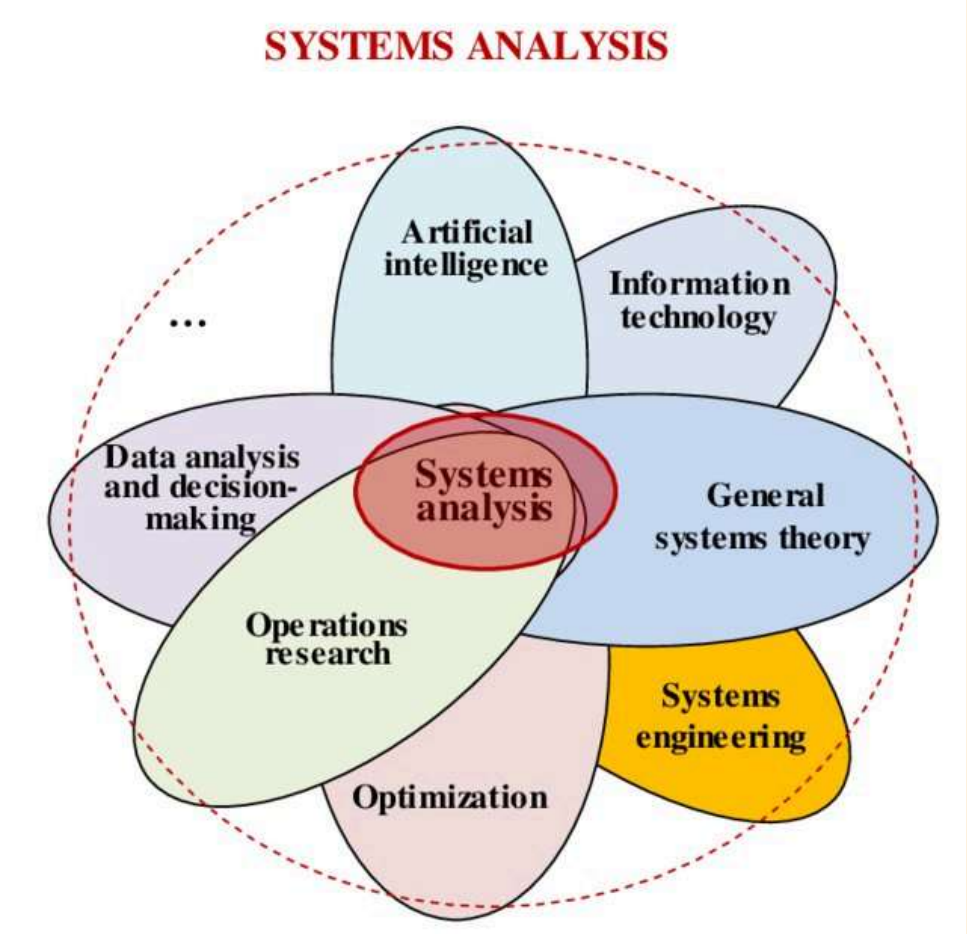
LEAN SIX SIGMA

a methodology for reducing process costs and increasing process quality (to improve processes)

USER INVOLVEMENT

getting user input into process spec and design (to improve the process specs and designs)

DIAGRAM OF SYSTEMS ANALYSIS



(Source: Dmitry A. Novikov, https://www.researchgate.net/publication/300131568_Systems_Theory_and_Systems_Analysis_Syste lo=1)

EXAMPLE OF AN IT PROCESS PROBLEM

WEBSITES FOR THIS COURSE

What things have gone wrong (and what were the fixes)?

Lets list a few

Systems analysis should shed light on fixing such things

So let's systems analyze it!

WHICH SYSTEMS ANALYSIS TECHNIQUES MIGHT WORK? WHY?

REVIEW OF SOME SYSTEMS ANALYSIS APPROACHES

Let us discuss the applicability of each

Suggestions: class discussion, small group discussions, or chat window

PROCESS MAPPING

visualizations of existing processes (to identify possible improvements)

ROOT CAUSE ANALYSIS

finding underlying causes of problems (to identify what to improve)

BENCHMARKING

comparing organizational processes to other organizations' processes (to identify areas to improve)

PROTOTYPING

building throwaway test solutions to decide what to build for real (to identify how to do improvements)

COST-BENEFIT ANALYSIS

comparing the benefits of a process change to its costs (to decide if improvement is worth doing)

AUTOMATION

using technology to boost labor productivity (to improve process quality and cost)

LEAN SIX SIGMA

a methodology for reducing process costs and increasing process quality (to improve processes)

USER INVOLVEMENT

getting user input into process spec and design (to improve the process specs and designs)

WHICH OF THESE ARE APPLICABLE TO OUR PROBLEM?

FISHBONE DIAGRAMS (ISHIKAWA DIAGRAMS)

They look like a fish skeleton (well, sort of)

They are for understanding root causes of a problem

They identify and can categorize potential causes

SHAPED LIKE A SKELETON (SEE IMAGES & TEMPLATE)

The fish head is for the problem

The fish bones are for the causes

SOME STEPS IN MAKING FISHBONE DIAGRAMS

1. Write problem at head of diagram
2. Draw a branch from the spine for each category of possible causes

environmental, staffing, mechanical, ..., it varies (ASQ, Wikipedia, etc.)

3. From each branch draw branchlets, one for each possible cause
4. From branchlets draw twigs for more specific "sub"-causes
5. As needed, sub-twigs may also be added for even more specific causes

WHY AND WHEN TO USE FISHBONE DIAGRAMS?

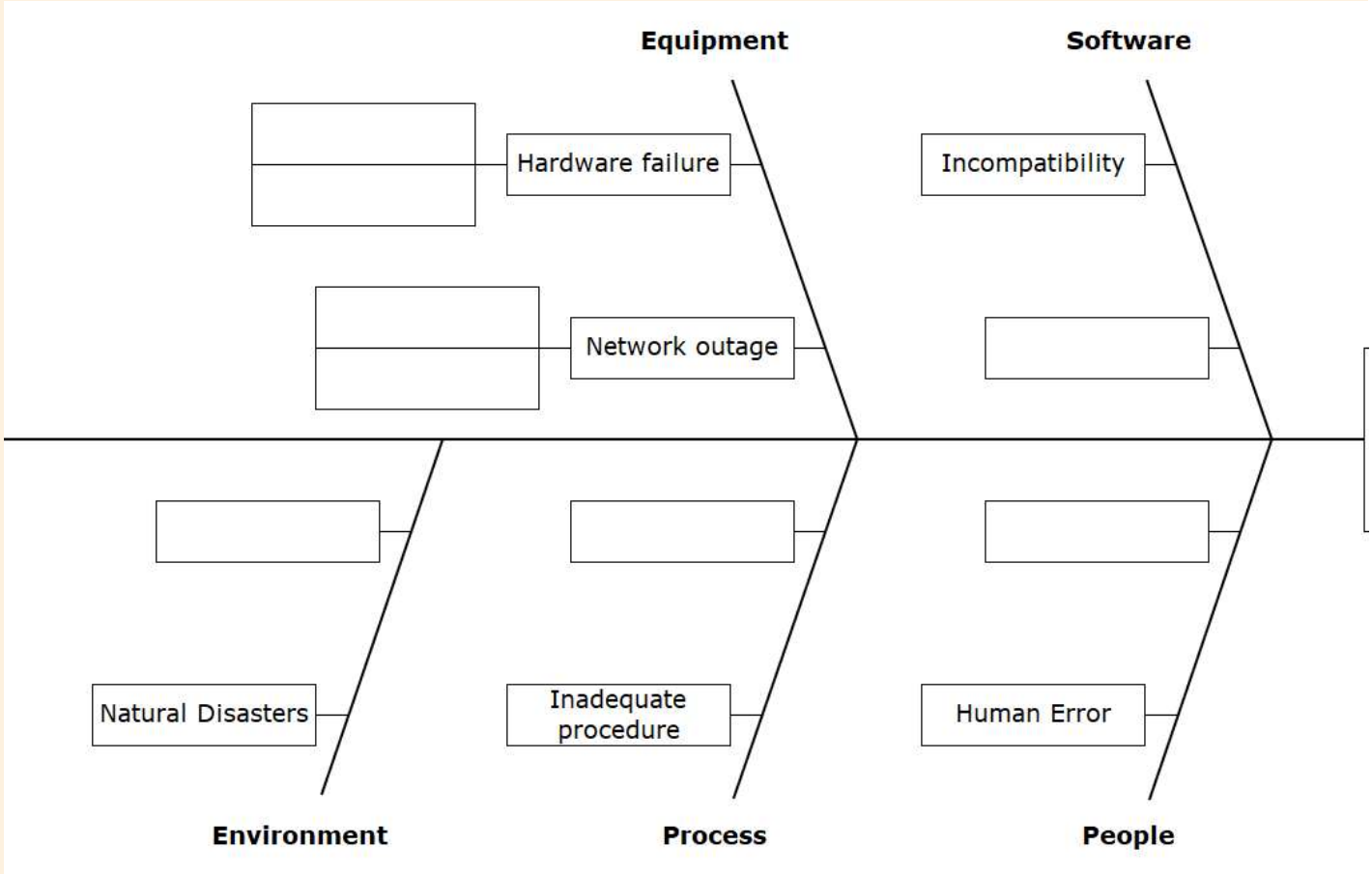
Structured, step by step approach for identifying causes
Help to see patterns and connections within the space of possible causes
The visualization helps communicate possible causes within and outside the group
The point is to support improvement in the processes under consideration

FIND A USEFUL OR INTERESTING FACT ABOUT FISHBONE DIAGRAMS

LET'S PUT THE THEORY INTO PRACTICE NOW! TOGETHER, LET'S FILL OUT THE FISHBONE DIAGRAM BELOW.

FISHBONE DIAGRAMS — ANOTHER EXAMPLE

(with blanks to fill in some of)



(Provided by Peng-Hung Tsai)

MORE EXAMPLE APPLICATIONS OF FISHBONE DIAGRAMS

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF PRODUCT DEFECTS

Issues identified might be materials, human error, machine breakdown, QC, ... any others?

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF SLOW PROJECT PROGRESS

Issues identified might be ... what?

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF DECLINING SALES

Issues identified might be ... what?

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF INFINITYFREE LOSING WORK

Issues identified might be ... what?

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF SLOW WEB PAGE LOADING

Issues identified might be ... what?

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF LOWERED STUDENT LEARNING

Issues identified might be ... what?

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF MEDICAL PATIENT RETURNING TO DOCTOR/HOSPITAL

Issues identified might be ... what?

INVESTIGATE (CHOOSE ONE: CAUSES, RESULTS, COSTS, SOLUTIONS, PROCESS, REDUCTION) OF WORKPLACE INJURIES

Issues identified might be ... what?